🔑 **Answer Key — Teacher Copy · Lesson 5-9**

MSTAR PRACTICE WORKSHEET

Lesson 5-9: Area of Regular Polygons

Grade 6 Mathematics · Standard 6.GR.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

A regular hexagon is divided into 6 equal triangles from the center. Each triangle has a base of 6 ft and a height of 5.2 ft. What is the area of one triangle?

- (A) 11.2 sq ft
- (B) 15.6 sq ft
- (C) 15.6 ft
- (D) 31.2 sq ft

$A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 6 \times 5.2 = 15.6$ square feet.

Question 2 SELECTED RESPONSE — CORRECT: C

Using the triangle from the previous question, what is the total area of the hexagonal skylight?

- (A) 15.6 sq ft
- (B) 62.4 sq ft
- (C) 93.6 sq ft
- (D) 187.2 sq ft

Total area = 6 triangles × 15.6 sq ft = 93.6 square feet.

Question 3 SELECTED RESPONSE — CORRECT: C

A regular hexagon is split into 6 triangles, each with base 4 cm and height 3.5 cm. What is the total area?

- A 14 sq cm
- B 21 sq cm
- C 42 sq cm
- D 84 sq cm

> Each triangle: ½ × 4 × 3.5 = 7 sq cm. Total: 6 × 7 = 42 sq cm.

Question 4 SELECTED RESPONSE — CORRECT: B

How many triangles can you make from the center of a regular pentagon?

- A 3
- B 5
- C 6
- D 10

> A regular pentagon has 5 sides, so drawing lines from the center to each vertex creates 5 equal triangles.

Question 5 SELECTED RESPONSE — CORRECT: D

A regular hexagonal floor tile splits from its center into 6 congruent triangles, each with an area of 12 cm^2. What is the total area of the tile?

- A 12 cm^2
- B 18 cm^2
- C 36 cm^2
- D 72 cm^2

> A hexagon has 6 sides, so it splits into 6 equal triangles. Total = 6 × 12 = 72 cm^2.

Question 6 SELECTED RESPONSE — CORRECT: C

A regular pentagonal tile splits into 5 congruent triangles. Each triangle has a base of 4 in and a height of 6 in. What is the total area of the tile?

- A $12\ in^2$
- B $24\ in^2$
- C $60\ in^2$
- D $120\ in^2$

One triangle = $\frac{1}{2} \times 4 \times 6 = 12\ in^2$. A pentagon has 5 triangles, so total = $5 \times 12 = 60\ in^2$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

A regular hexagon is decomposed from its center into 6 identical triangles. Each triangle has a base of 8 cm and a height of 7 cm. What is the area of the hexagon?

- (A) 28 square centimeters
- (B) 56 square centimeters
- (C) 168 square centimeters
- (D) 336 square centimeters

> Each triangle = ½ × 8 × 7 = 28 square centimeters. Six of them: 6 × 28 = 168 square centimeters.

- **A:** That is ONE triangle. The hexagon is made of six of them.
- **B:** That is two triangles. All six make up the hexagon.
- **D:** The triangles were never halved. Each is half of 8 times 7, which is 28, and six of those give 168.

Part B — correct answer: C

Why can a regular polygon be decomposed into identical triangles from its center?

- (A) Every polygon can be split into identical triangles.
- (B) Triangles are the only shape with a known area formula.
- (C) All its sides are equal and all its angles are equal, so each center-to-side triangle is congruent.
- (D) The center of a polygon is always a vertex.

> Regular means equal sides and equal angles, so the triangles formed from the center to each side are congruent and share the same height.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL statements that are true about the area of a regular polygon:

- A. It equals the number of sides times the area of one center triangle.
- B. Each center triangle has the same height for a regular polygon.
- C. Only regular polygons with an even number of sides can be decomposed this way.
- D. A regular polygon has all sides equal and all angles equal.
- E. The height of each center triangle is measured perpendicular to a side.

> A, B, D, and E are true. C is false — a regular pentagon has 5 sides and decomposes into 5 congruent triangles just as well.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A regular hexagon and a regular pentagon both have triangles with the same base (6 ft) and the same height (5 ft). Which polygon has the greater total area? Explain why the number of sides matters.

> **Model answer:** One triangle is $\frac{1}{2} \times 6 \times 5 = 15$ sq ft in both polygons, so the only difference is how many copies each shape holds: the hexagon has 6 for 90 sq ft and the pentagon has 5 for 75 sq ft. The hexagon has the greater area because the number of sides is exactly the number of identical triangles you get to add.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
